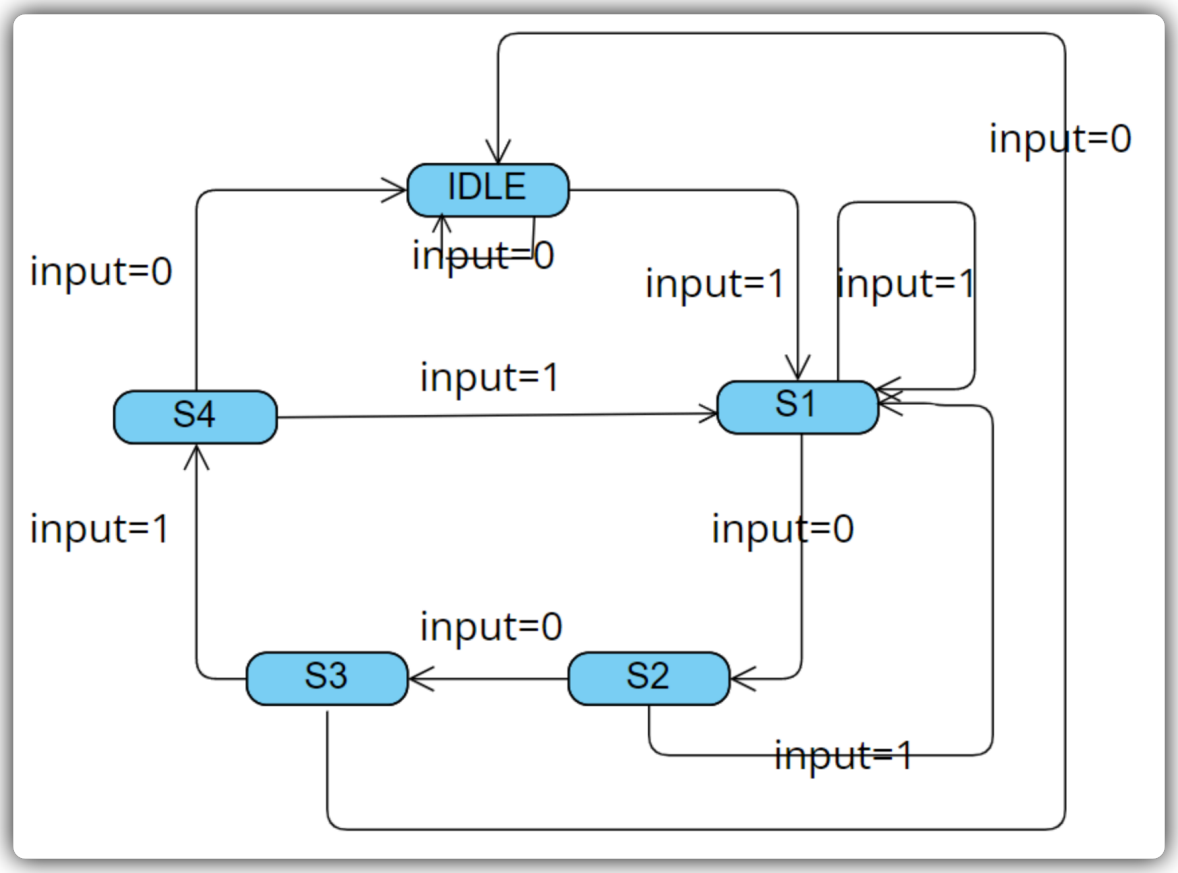
module 05: 序列检测

1.序列检测

- 使用verilog代码，设计电路，检测序列为10110，检测到的时候输出1，没检测到的时候输出0

2.序列检测原理



3.RTL

```
module seq_detector(
  input  x,
  input  clk,
  input  reset,
  output reg z
);

  reg[2:0] state, nextstate;
  parameter s0 = 3'b000, s1 = 3'b001, s2 = 3'b010, s3 = 3'b011, s4 = 3'b100;

  always@(posedge clk or negedge reset) begin
    if(!reset) begin
      state <= s0;
```

```
      nextstate <= s0;
    end
    else begin
      state <= nextstate;
    end
  end

  always@(*) begin
    case(state)
      s0: if(x) nextstate = s1; else nextstate = s0;
      s1: if(x) nextstate = s1; else nextstate = s2;
      s2: if(x) nextstate = s3; else nextstate = s0;
      s3: if(x) nextstate = s4; else nextstate = s2;
      s4: if(x) nextstate = s1; else nextstate = s2;
      default: nextstate = s0;
    endcase
  end

  always@(posedge clk or negedge reset) begin
    case(state)
      s4: if(x) z <= 1'b0; else z <= 1'b1;
      default: z <= 1'b0;
    endcase
  end

endmodule
```